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Efficient topic identification for urgent MOOC Forum posts using BERTopic and traditional topic modeling techniques

Nabila Khodeir[1] · **Fatma Elghannam**[1]

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Abstract
MOOC platforms provide a means of communication through forums, allowing learners to express their difficulties and challenges while studying various courses. Within these forums, some posts require urgent attention from instructors. Failing to respond promptly to these posts can contribute to higher dropout rates and lower course completion rates. While existing research primarily focuses on identifying urgent posts through various classification techniques, it has not adequately addressed the underlying reasons behind them. This research aims to delve into these reasons and assess the extent to which they vary. By understanding the root causes of urgency, instructors can effectively address these issues and provide appropriate support and solutions. BERTopic utilizes the advanced language capabilities of transformer models and represents an advanced approach in topic modeling. In this study, a comparison was conducted to evaluate the performance of BERTopic in topic modeling on MOOCs discussion forums, alongside traditional topic models such as LDA, LSI, and NMF. The experimental results revealed that the NMF and BERTopic models outperformed the other models. Specifically, the NMF model demonstrated superior performance when a lower number of topics was required, whereas the BERTopic model excelled in generating topics with higher coherence when a larger number of topics was needed.The results considering all urgent posts from the dataset were as follows: Optimal number of topics is 6 for NMF and 50 for BERTopic; coherence scores is 0.66 for NMF and 0.616 for BERTopic; and IRBO scores is 1 for both models. This highlights the BERTopic model capability to distinguish and extract diverse topics comprehensively and coherently, aiding in the identification of various reasons behind MOOC Forum posts.

Keywords Educational technology · Online learning · Student engagement · MOOCs · Topic modelling · Urgent posts · BERTopic · Machine learning

Extended author information available on the last page of the article

1 Introduction

Online education has the ability to cater to a wide range of learning preferences and styles. Multimedia-rich course materials, interactive simulations, and personalized feedback mechanisms can enhance the learning experience and foster deeper engagement (Dhawan 2020). This diversity has enabled educational institutions to better meet the diverse needs of their students, even in the face of many challenges (Hodges et al. 2020). In addition, accessibility has been a major factor in the growing importance of online education. By transcending geographical barriers, online platforms have democratized education, enabling learners from diverse backgrounds to access high-quality educational resources regardless of their physical location (Zawacki-Richter 2021). This has proven to be particularly beneficial for individuals with disabilities, caregiving responsibilities, or those living in remote or underserved areas, who may have faced significant barriers in traditional educational settings (Hodges et al. 2020). The COVID-19 crisis has also accelerated this trend as it has thrust online learning into the spotlight as a vital component of modern academia (Bates 2015). This forced educational institutions to quickly adapt and adopt distance teaching methods (Dhawan 2020). This rapid transformation has underscored the inherent advantages of online education, which include increased accessibility, flexibility, and the ability to accommodate diverse learning styles (Bozkurt and Sharma 2020).

Massive Open Online Courses (MOOCs) have emerged as a significant educational advancement, gaining widespread adoption across various institutions worldwide. In these courses, students engage with pre-designed materials and undertake independent study as a major component of their learning process. However, interaction plays a pivotal role in online learning, including MOOCs, ensuring the quality of education (Trentin 2000).

To enhance students' learning experiences and performance, monitoring their interactions and addressing their requirements effectively is crucial. MOOCs provide forums facilitating communication between learners and instructors, enabling them to engage in discussions and seek support (Alyssa Friend Wise and Cui 2018) However, the effectiveness of traditional student-to-instructor communication faces challenges in the MOOC context due to the sheer volume of students involved and the diverse range of learner backgrounds, needs, and objectives (Jacobsen 2019).

Notably, while learners generate a substantial number of posts throughout the course, only around one-fifth of these posts are considered urgent and require immediate attention from the instructor (Akshay Agrawal et al. 2015).

To deliver a high-quality learning experience, it is essential to prioritize the identification and analysis of learners' urgent requests (posts) and promptly address them by notifying the relevant department. Previous studies have categorized learners' requirements into urgent and non-urgent categories (Almatrafi, Johri, and Rangwala 2018; Khodeir 2021; El-Rashidy et al. 2024). This analysis can aid instructors in managing many posts and prioritizing their answers.

However, tackling the issue only as a classification issue without considering the context of the learners' requests limits the ability to accurately identify the specifics of the request, which in turn assists in taking the necessary action in this respect. Urgent posts within the discussion forums can cover various topics and issues. Learners may seek immediate assistance for problems, clarification on course materials, guidance on assignments or exams, or even express concerns about technical difficulties. Urgent posts can also be related to administrative matters such as registration issues, grading problems, or questions about course policies.

There is a growing demand for automated techniques to analyse and understand these urgent posts' content to address this challenge. Natural Language Processing (NLP) and machine learning techniques can automatically identify and categorize urgent posts based on their topics and content (Alrajhi and Cristea 2023; Zankadi et al. 2023). By leveraging these techniques, administrators can efficiently prioritize and respond to urgent posts, ensuring learners' needs are met and issues are resolved promptly.

Topic modelling uses unsupervised machine learning techniques. Without labeled data, topics are identified by counting words and grouping similar word patterns. Topic models can group similar documents and build new connections between topics, which can assist in organizing and provide insights for understanding a large corpus (Abdelrazek et al. 2023). Compared to supervised algorithms, it requires less manual effort because it does not need to be trained on data manually labelled by humans.

Topic modelling is defined as finding topics, as weighted sets of words that best represent the information of the documents (Angelov 2020). Several models are used to create such topics. Generative probabilistic models, such as Latent Dirichlet allocation (LDA) (David M Blei, Ng, and Jordan 2003; D. Blei 2001), Latent Semantic Indexing (LSI) (Landauer, Foltz, and Laham 1998; Landauer et al. 2013), and Non-Negative Matrix Factorization (NMF) (D. Lee and Seung 2000; Xu, X. Liu, and Gong 2003) are well-known traditional approaches to perform topic modeling.

Latent Dirichlet Allocation LDA is the most famous method for topic modeling. LDA is an unsupervised clustering technique proposed by Blei et al. (David M Blei, Ng, and Jordan 2003) used for text analysis. It is a generative probabilistic model in which each document is considered a distribution of topics, and each topic is considered a distribution of words (Guo and J. Li 2021). The LDA model has two hyperparameters: Alpha (α) and Beta (β). The distribution of document topics is controlled by (α), while the distribution of topic words is controlled by (β). All MLbased /or LDA approaches fundamentally treat the corpus as a bag of words to get the initial document term matrix, which means it does not consider word order, synonyms, syntactic information, the semantics of words, etc. Despite their usefulness in text analysis, LDA models have a notable disadvantage: they can produce ambiguous or inconsistent topics, particularly when dealing with noisy, sparse, or diverse data (Jelodar et al. 2019).

Latent Semantic Indexing (LSI) is a traditional topic modeling technique used to explore relationships between a large set of documents and the terms they contain. It identifies word usage patterns across the documents and grouping terms with similar patterns (Dumais 1994; Kontostathis and Pottenger 2006).

Non-Negative Matrix Factorization (NMF) (D. D. Lee and Seung 1999) is popular unsupervised topic modeling technique for reducing the dimension of the input corpora. It uses the factor analysis method to give the words with less coherence comparatively less weight (D. Lee and Seung 2000), and it has been widely used to discover the underlying relationships between texts and identify latent topics (Arora et al. 2012).

BERT, Bidirectional Encoder Representations from Transformers (BERT) from Google (Devlin et al. 2018), has become the focus in recent years for achieving state-of-the-art performance in several NLP tasks (Grootendorst 2022). The BERT embeddings model provides an improvement for the context dependency issue. This model is context-sensitive, with each word containing a variety of embeddings depending on where it appears in the sentence. It achieves good results in generating contextual word and sentence vector representations. The impressive results of Bert's pretrained language representations inspired the creation of BERTopic. BERTopic is an end-to-end topic modeling tool that utilizes BERT embeddings and a class-based term frequency-inverse document frequency (c-TF-IDF) to create dense clusters. In contrast to the traditional document-level TF-IDF technique, the class-based TF-IDF procedure models the importance of words in clusters rather than individual documents. This makes it possible to produce topic-word distributions for every cluster of documents (Grootendorst 2022). BERTopic technique creates topic representations based on four key components: First, embedding, which involves con- verting documents into their embedding representations using the previously trained language model. Second, to improve the clustering process, the dimensionality of the generated embedding is reduced before clustering the documents using the Uniform Manifold Approximation and Projection (UMAP) technique. Third, the unsupervised machine learning clustering technique HDBSCAN. Finally, topic representations are obtained from the documents using a class-based TF-IDF (c-TF-IDF) variant.

This paper introduces an urgent students' posts Topic Model (UPTM) based on Natural Language Processing (NLP) and the BERTopic modelling technique. We aim to identify and detect the cause of urgent posts based on spontaneous learners' interactions in MOOCs discussion forums.

2 Previous work

Massive Open Online Courses (MOOCs) have become an increasingly popular mode of delivering higher education to large, geographically dispersed audiences (Chen 2014). A key component of the MOOC experience is the online discussion forums, which serve as a hub for student-student and student-instructor interactions (Kizilcec, Piech, and Schneider 2013).

Prior research has highlighted the importance of these forums in fostering a sense of community and social presence for MOOC learners, which in turn can positively impact student engagement, satisfaction, and learning outcomes (Wang and Baker 2018). For example, a study by (Gillani and Eynon 2014) found that students who actively participated in MOOC forums were more likely to complete the course and achieve better learning outcomes compared to their less engaged peers.

Furthermore, (Wang and Baker 2018) examined the relationship between student interactions in MOOC forums and their perceived sense of community and social presence. Their findings suggest that the quality and quantity of student-to-student and student-to-instructor interactions in the forums are key predictors of learners' feelings of connectedness and belonging within the MOOC environment.

Recently (Wei et al. 2023) explored engagement strategies within MOOC forums, with a focus on low-achieving students. The findings revealed that low-scoring students who made significant progress tended to engage less in peer-peer and peer-teacher interactions. Instead, learner-content interactions emerged as crucial for advancing learning, acquiring new information, and enhancing linguistic knowledge. Quick feed- back and targeted suggestions were preferred over extensive peer discussions. The implications from this study underscore the importance of continuous teacher support and encouragement of peer-generated content.

Furthermore, (Bonafini et al. 2017) found that increased engagement in both forums and video content correlated positively with higher course achievement. Interestingly, the intention to certify played a moderating role in video consumption and overall achievement. These insights underscore the critical role that MOOC discussion forums play in shaping the overall learning experience and student success in these online learning platforms (Kizilcec, Piech, and Schneider 2013).

NLP holds immense potential for enhancing education, and its application in MOOCs continues to evolve. Researchers and practitioners alike are exploring novel ways to leverage NLP for better learning outcomes. NLP plays a crucial role in personalizing instructional materials for individual students. By analyzing web content, it tailors learning resources to student interests. Moreover, NLP assists in generating test questions for teachers and semi-automating educational technology systems. Recent phenomena like MOOCs have opened up new research opportunities, bridging the gap between NLP and education (C. Li and Xing 2021; Wu et al. 2024). Furthermore, Advanced Natural Language Processing (NLP) significantly contributes to the analysis of educational data. Researchers have delved into diverse aspects, encompassing sentiment annotations, entity annotations, text summarization, and topic modeling (Shaik et al. 2022).

In MOOCs, topic modeling has been used to analyze student behavior, identify popular topics, and improve course design. Different studies utilized topic modeling to identify topics of interest based on MOOCs reviews. The authors of (Peng et al. 2016) suggested a Like-Latent Dirichlet Allocation (Like-LDA) model that uses the behavioral characteristic "like" to create a profile of learner topic interest. (S. Liu et al. 2017) Suggested an author-topic model based on an unsupervised learning concept to extract learning topics for each learner. Their study aimed to analyze and focus on learners' interests to facilitate further personalized course recommendations based on topic information hidden in the review data in MOOCs.

The study of (Vytasek, Alyssa F Wise, and Woloshen 2017) explores four potential methods for using topic modeling to help instructors manage and organize MOOCs discussion forums. Different approaches were proposed for topic modeling, from tra- ditional algorithms to the most recent deep learning-based methods. (Ramesh et al. 2014) built a topic model using LDA to detect latent topics in students' writings in forum discussions to predict student retention in MOOCs. The authors (Lubis, Rosmansyah, and Supangkat 2019), developed a topic model using LDA, sentiment analysis and a sentence filtering approach based on helpful subjective posts in MOOCs. The work in (B. Yang et al. 2022) used latent semantic analysis (LSI) to classify and identify learners with distinct longitudinal profiles of topic-relevant forum posting in MOOCs. The work of (Bafna and Saini 2020) compared various topic modeling techniques including LDA, LSI, and NMF on MOOCs discussion forums data. (B. Yang et al. 2022) proposed a hierarchical NMF-based framework for topic modeling in MOOCs. Their framework not only identified topics at different levels of granularity but also helped to understand the relationships between topics.

Pre-trained word embeddings have dominated the semantic representation space since they were first developed. Pre-trained word embeddings, such as Word2vec and GloVe, on the other hand, compute a single static representation for each word. This, in turn, results in disregarding the significance of context in illuminating the precise meanings of words (Camacho-Collados and Pilehvar 2020). Most recent research in natural language processing has been interested in pre-trained transformer-based lan- guage models like BERTs (Terragni et al. 2021; Zhao et al. 2021). (Bianchi, Terragni, and Hovy 2020) suggested using Sentence-BERT to generate the document embed- ding vectors instead of bag-of-words (BoW) representations. Their results showed that context document embeddings produce significantly more coherent topics than BoW representations. Mueller and Dredze extended this approach (Mueller and Dredze 2021) who also considered fine-tuning the representations. (Hoyle, Goel, and Resnik 2020) follow a similar direction and use knowledge distillation to combine neural topic models and pre-trained transformers (Zhou and Wakabayashi 2022). Proposed a jointly fine-tuned BERT model for phrase and sentence embeddings and utilized it on a phrase-level topic model. According to their experiments, the model could generate well-performed phrase-level topics.

Recently, a growing interest has been in leveraging BERT-based models for topic modeling in MOOCs. These research works utilize BERT's contextual embeddings to improve the accuracy and effectiveness of topic extraction from MOOCs text data. (Najmani et al. 2023) utilized Topic modeling to identify subjects that may interest learners within MOOC courses and, more broadly, to uncover latent themes within unstructured text collections without needing specific training data. BERTtopic were used in the work of (Zankadi et al. 2023) to extract topics of interest from the text content submitted by learners to enrich their course preferences in MOOCs. The authors (ibid.) extract the topical interest from the text content shared by learners on social media to enrich their course preferences in MOOCs. They applied NLP and topic modeling techniques using different traditional topic models approaches and BERTopic. Their results have demonstrated that BERTopic outperformed the other models. Other studies have explored hierarchical topic modeling frameworks, such as (Meng et al. 2020), and joint models that combine document classification and topic modeling in MOOC contexts.

3 Topic modeling for urgent posts

The nature of urgent posts requiring instructors' attention can vary depending on learners' specific challenges. These challenges can be specific to the course, such as difficulties in understanding certain materials, or they can encompass broader issues like logistical or technological problems related to downloading videos, accessing YouTube and, file loading. Recognizing the topics associated with urgent posts allows instructors to address them with greater specialization and professionalism.

For this study, a range of topic modeling techniques, including LDA, LSI, NMF, and BERTopic, were employed to identify the topics discussed in learners' urgent posts. Utilizing these approaches aimed to gain insights into the underlying themes and subjects that emerge from the urgent posts. This information can inform instructors' responses and interventions, enabling them to support the learners better.

A series of preprocessing techniques are applied to prepare the textual data from the learners' posts. Subsequently, feature extraction is done on the data using LDA, LSI, and NMF techniques. The prepared textual data is then used to train four topic modeling techniques: LDA, LSA, NMF, and BERTopic. The performance of these techniques is evaluated using topic evaluation metrics, such as Coherence metrics score and Inverted Rank-Biased Overlap (RBO) score. The following sections provide comprehensive details about the data and the step-by-step process of extracting topics using automated topic modeling approaches.

3.1 Data set

We conducted our experiments using the Stanford MOOC Posts dataset (A. Agrawal and Paepcke 2014), which consists of 29,590 learner forum posts from eleven Stanford University public online courses. The dataset encompasses courses from three domains, namely Education, Humanities, and Medicine, with an equal distribution of courses from each domain. Table 1 illustrates the distribution of courses within each domain.

Table 1 Stanford MOOC courses and their course types

Course Type	Course display name	counts
Education	Education/EDUC115N/How_to_Learn_Math	9879
Humanities	GlobalHealth/WomensHealth/Winter2014	2141
	HumanitiesScience/StatLearning/Winter2014	3029
	HumanitiesScience/Stats216/Winter2014	328
	HumanitiesSciences/EP101/Environmental_Physiology	2467
	HumanitiesSciences/Econ-1/Summer2014	1583
	HumanitiesSciences/Econ1V/Summer2014	160
Medicine	Medicine/HRP258/Statistics_in_Medicine	3321
	Medicine/MedStats/Summer2014	1218
	Medicine/SURG210/Managing_Emergencies_What_Eve...	279
	Medicine/SciWrite/Fall2013	5184

Table 2 presents a sample of frames from the Stanford MOOCs posts dataset, showcasing manual labeling that includes urgency scores for different types of courses. The figure highlights the diverse causes of urgency for the posts. For instance, the last post's urgency is attributed to quiz issues, while the first and second posts are related to problems with downloading videos and accessing YouTube. This variation in the topics or causes of students' urgent posts underscores the importance of topic modeling in identifying and directing these topics to the relevant individuals responsible for addressing such issues.

The Stanford MOOCs Posts dataset has undergone manual labeling, encompassing various dimensions, one of which is urgency. The urgency dimension is rated on a scale ranging from 1 to 7. A post assigned a rank of 1 indicates that it is not urgent and does not require an immediate response from the instructor. On the other hand, a post assigned a rank of 7 signifies that it is of utmost importance and demands the instructor's immediate attention.

For this study, only posts with a rank above four are considered urgent and included in the dataset for model training. After applying this criterion, a total of 6,415 posts were obtained. Table 3 provides an overview of the distribution of posts across different types of courses.

By employing this binary classification approach, it is found that urgent posts account for approximately 20% of all posts across the various domains. This relatively small percentage of critical cases significantly reduces the instructor's workload, as they only need to focus on tracking and responding promptly to the urgent posts, while the non-urgent posts make up the remaining 80% of the workload.

To assess the effectiveness of various topic modeling techniques, we will utilize students' posts labeled as urgent as our reference datasets. These urgent posts are distributed across different types of courses, necessitating experiments on different subsets of the dataset. The dataset will be divided into four distinct groups, namely Group A, Group B, Group C, and Group D, based on the course types.

- Group A includes all urgent posts from the dataset, regardless of the course type.
- Group B includes only urgent posts from the Education course type.
- Group C includes only urgent posts from the Humanities course type.
- Group D includes only urgent posts from the Medicine course type.

By partitioning the dataset in this manner, we can evaluate the performance of the topic modeling techniques on specific subsets of urgent posts, allowing for a more targeted analysis based on course types.

3.2 Data preprocessing

Text preprocessing is crucial for cleaning and preparing text data before feeding it into a model. Several preprocessing tasks are applied, including text cleaning, tokenization, stop word removal, and part-of-speech tagging.

The first step involves converting reductions, such as "won't" and "can't," into their natural forms to improve clarity. Next, the data is cleaned by removing links,

Table 2 Sample of frames for the Stanford MOOCs dataset

Text	Urgency	CourseType
I could download the videos at work but I cannot watch them there. I can watch them at home but I cannot download at home.	6	Humanities
YouTube is blocked from my country, if I can't not get access to the lectures, I am afraid I have to stop now. If you are the instructor or TA, can you give me some clarification of the video download issue, thank you.	7	Humanities
Hi, my name is <nameRedac<anon_screen_name_redacted>>. I live in Lithuania. I am here to learn something new for my free time activities and improve my english.	2	Humanities
Answers to QUIZ: PRICE CEILINGS AND FLOORS are wrong. The explanation disagrees with the marked answers. Please check!	7	Humanities

Table 3 Number of urgent posts along course types

Group A	Course Type	Counts
Group B	Education	461
Group C	Humanities	2474
Group D	Medicine	3480

numbers, emojis, special characters, punctuation, and excess spaces. Additionally, commonly used words like "a," "an," "the," and others are eliminated. The NLTK library utilizes a predefined list of stop words. The specific version of NLTK used in this context is 3.8.1. This list comprises frequently occurring words that typically carry little semantic weight and can be safely removed from the text.

In addition, all characters in the text were converted to lowercase. NLTK package contains classes and interfaces for part-of-speech tagging, or simply “tagging”. NLTK tagging was employed, and only nouns, adjectives, verbs, and adverbs were retained. Subsequently, the text underwent tokenization, splitting the sentences into individual words.

3.3 Experimental setup

After the preprocessing phase, the feature extraction stage was carried out. In this stage, the data needs to be converted into a numerical form that can be understood by machine learning algorithms. This involves transforming the text documents into vector representations, where each post represents one document. The transformation process can be classified into two categories based on the adopted topic modeling techniques: Bag of Words transformation and sentence transformer embedding model. These techniques enable converting textual data into numerical representations suitable for further analysis and modeling.

The Bag of Words transformation is a feature extraction technique for the LDA, LSI, and NMF models. It represents a document data model that focuses on the occurrence of words within a document (Diera et al. 2022). A dictionary and a corpus need to be created to train these models. The dictionary captures the mapping between normalized words and their assigned integer IDs, while the corpus lists tuples for each word ID and its frequency within the document.

In natural language processing, there are two types of hyperparameters: latent topics and Dirichlet priors. The LDA, NMF, and LSI models utilize latent topics as hyperparameters. On the other hand, Dirichlet priors (alpha and beta) are specifically used for the LDA model. Typically, the alpha and beta values are set to "Auto," allowing the model to learn the optimal values for these parameters during execution.

Determining the number of latent topics before running the LDA, NMF, and LSI models is important. A grid search approach was employed to find the optimal number of topics for these models, and this process will be detailed in the evaluation section. For BERTopic, the sentence transformer embedding model was utilized to convert the text data into numerical representations. Specifically, the "all-MiniLM-L6-v2" sentence-transformers model, designed for semantic similarity tasks, was employed.

This model has demonstrated strong performance across various applications.

To further process the numerical representations, the UMAP algorithm was employed. UMAP is the default dimensionality reduction technique used in BERTopic. It helps reduce the dimensionality of the numerical representations, making them more manageable and suitable for subsequent analysis.

Reducing the dimensionality of a dataset is crucial for identifying groups of documents with similar semantic properties. This reduction process helps preserve the dataset's local and overall structure. In the UMAP dimensionality reduction technique, one of the hyperparameters is the number of nearby "n_neighbors" sample points used in the manifold approximation. Choosing smaller values for this parameter provides a more localized perspective of the embedding structure, while larger values offer a

To optimize the coherence values, different values for the "n_neighbors" parameter were tested. Setting it to 0.7 yielded the highest coherence values, indicating an optimal balance for the dataset. Another important hyperparameter in UMAP is "n_components," which defines the reduced dimensions of the embeddings. The default value for "n_components" is set to 5, aiming to minimize dimensionality while maximizing the information captured in the generated embeddings. Genrally we we optimized hyperparameters such as UMAP's n_neighbors and BERTopic's n_components by testing different values against coherence scores. We selected the values that achieved the highest coherence scores.

The remaining UMAP parameters were set to their default values. "calculate probabilities" was set to "True", determining the likelihood that a document belongs to any given topic. The number of topics was set to "auto," allowing the algorithm to determine the optimal number automatically. The minimum distance parameter was set to 0.05, indicating the minimum acceptable separation between points in the reduced-dimensional space. The metric used for distance calculation was set to "cosine," which measures the angular similarity between vectors. The random state was set to "100" to ensure reproducibility of the results.

Two important components in BERTopic that are often overlooked are the CountVectorizer and cTFIDF computation. These components play a crucial role in constructing the topic representations and offer flexibility for customization. The CountVectorizer has various settings that can be adjusted to enhance the quality of the topic representations. For example, the "ngram_range" option allows choosing the number of tokens in each entity within a topic representation.

In BERTopic, TFIDF is modified to work on a cluster/categorical/topic level rather than a document level. This modified version, cTFIDF, considers the distinctions between documents in different clusters, providing a more accurate representation of topics derived from the bagofwords matrix.

3.4 Topic identification with BERTopic Model

BERTopic is an advanced approach to topic modeling that incorporates BERT embeddings and c-TF-IDF. It produces concise and well-defined clusters,

Table 4 BERTopic modelh´yperparameters

calculate_probabilities	ngram_range	**nr_topics**
True	(1,3)	auto

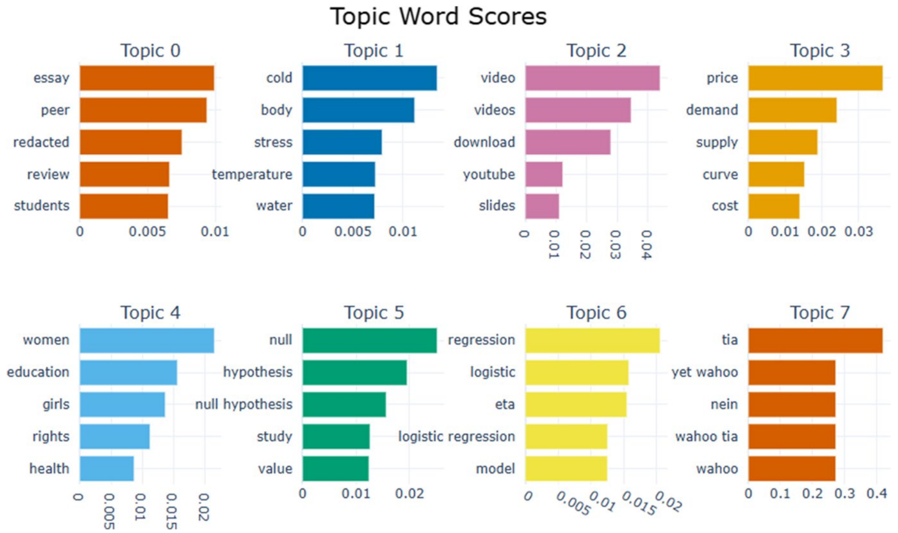

Fig. 1 Part of topic word scores for dataset group A

making it easier to interpret the topics. BERTopic ensures that important words are preserved in the topic descriptions, enhancing the quality of the generated topics. One of the key features of BERTopic is its ability to extract coherent topic representations. It achieves this by utilizing a modified version of TF-IDF, known as c-TF-IDF, which considers class information and improves the clustering process.

BERTopic also automatically determines the number of topics, saving the user from manually specifying this parameter. It provides convenient visualization options to explore and understand the generated topics effectively. In addition, BERTopic offers support for various topic modeling variations, including class-based topic modeling, which considers class labels during the modeling process. It also enables hierarchical topic reduction, resulting in a more organized and structured representation of topics. Moving on to applying BERTopic in all dataset groups, let's focus on group A. Using the setting shown in Table 4, we obtained 57 automatically generated topics. Figure 1 provides a visualization of the topic word scores for the first eight topics within this group. Certain topics, like topics 1,3, and 4, are directly related to the course material. On the other hand, there are general topics, such as topics 0, 2, and 6, which cover subjects like essays, questions and technological issues, including downloading videos.

3.5 Practical Applications of Proposed Algorithms in Educational Contexts

To give concrete examples of how the proposed algorithms can be applied in educational settings. We used the trained BERTopic model to predict the main topics of urgent posts. Table 5 shows examples of the topics expected in learners' urgent posts. We can notice that topic 0 is related to the essay which is a general topic in all training courses. We also notice the different cases related to it, such as: How will articles be received, formatted, and calibrated. Topic 1 is associated with humanities and medicine course materials, covering discussions related to cold and temperature. The second topic primarily revolves around videos, downloads, and slides. It also en-compasses various related issues, such as very short videos, indicating video runtimes, and the necessity for downloadable videos in countries with blocked websites.

Such example highlighted the varying critical topics among students, emphasizing the need for specialized approaches to address them. There are two main categories: content-specific issues, which subject teachers address, and general problems like technical issues and Inquire and follow up on required assignments.

By addressing various critical aspects, we can save the teacher's time by focusing on subject-related concerns and topics that raise questions or cause confusion. General problems can be handled by specialists. Recognizing the cause of urgent forum posts and dealing with them through specialists allows for prompt feedback, which enhances students' motivation by providing timely recognition and guidance, fostering a sense of achievement and progress (Shanshan and Wenfei 2024). Additionally, it reduces anxiety by clarifying doubts and offering support, creating a more secure learning environment (ibid.). Moreover, prompt responses contribute to a sense of community within the course, encouraging interaction and engagement among peers and instructors (H.-H. Yang and Lin 2023). These psychological benefits are crucial for improving student retention and success in MOOCs.

3.6 Using BERTopic across different course domains

BERTopic offers the flexibility to apply class-based topic modeling, which we utilized in the context of different course types, namely Medicine, Education, and Humanities. Figure 2 illustrates the results of this analysis. Upon examination, it is noticeable that topics 0, 2, and 8 are general topics that appear across all training courses and are not specifically related to the course content. Conversely, topics related to the course content are not spread across multiple courses and are more distinct to each course type. This demonstrates how class-based BERTopic modeling can be applied across various course domains, effectively handling a wide range of subjects, languages, and cultural contexts.

To delve deeper into further analysis, we extended our investigation to the other dataset groups: B, C, and D. In group B, we obtained eight automatically generated topics, while group C yielded seventeen topics, and group D resulted in thirty-seven topics. For a visual representation of these topics, please refer to Figs. 3, 4, and 5,

Table 5 Predicted topics for some selected posts

Urgent Post	Predicted Topic	CourseType
hi sef going reiterate problem others trying grade peer responses error message comes enough calibration essays however currently essays submitted calibration essay differ student submission something needs done jo end enable calibration thanks nameredac anonscreen name redacted	0	Education
Why does all the sentences clump into one big paragraph after you submit the essay assignment in paragraph format?	0	Humanities
How will we receive the essays to peer review?	0	Medicine
I agree. In the video it was stated that his body temperature started to level out the longer he exercised and she didn't believe he would have gotten much colder. So if you kept moving I think it would prevent the temperature drop some what.	1	Humanities
Looks like there could be a technical glitch with the Cold Midterm.	1	Humanities
I think explanation given quiz question answers query difference high low exercise groups becomes more extreme adjusting age indicates effect exercise was obscured differences ages groups blood pressure increases age high exercise group be older low exercise group average raj	1	Medicine
I just started watching the course videos. All I'm finding are very short 2-4 minute videos that appear to be the home-work assignments. Am I missing something? Are there any 10-15 minute lectures on the site?	2	Education
I'm probably not putting this in the right place, but I don't see anything for technical issues. I'd like to request that the run-time of the videos be listed next to their names. This way we'll know at the beginning of the week, before playing each video, how much time we'll need to allot for each one. Thank you!	2	Humanities
hi I am student access internet china unknown ridiculous reasons lot useful websites are blocked here such youtube actually I have proxy even lecture videos other coursewares are not accessible introduction part I do not know be shows square block so is possible make things downloadable yes http connection similar is much easier more possible access proxy others such embedded videos I think be great course I not access be very sorry miss thank	2	Humanities
Agreed - also PowerPoint slides in a few places or Unit 2. Kind of messes up the whole problem.	2	Medicine

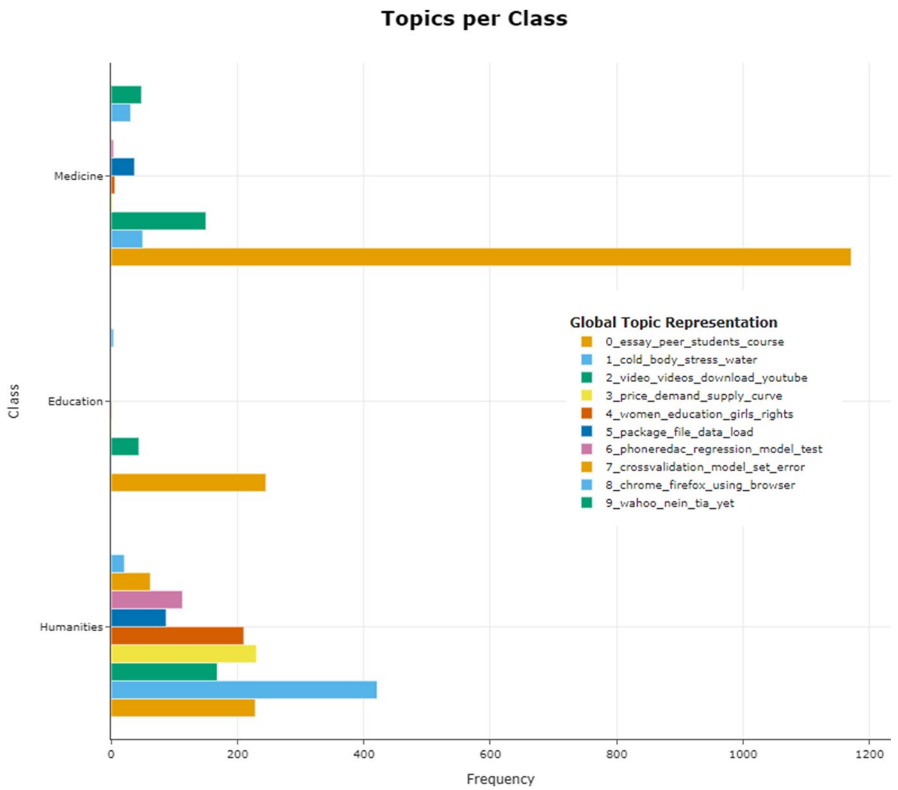

Fig. 2 Topics per course type

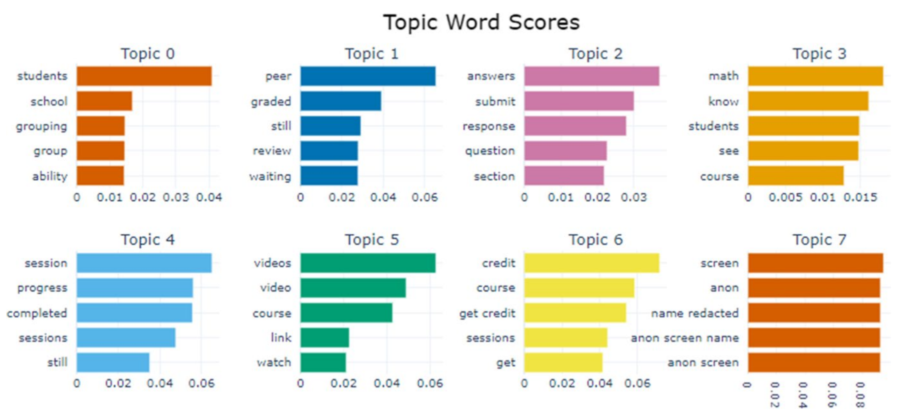

Fig. 3 Topic word scores for Dataset group B

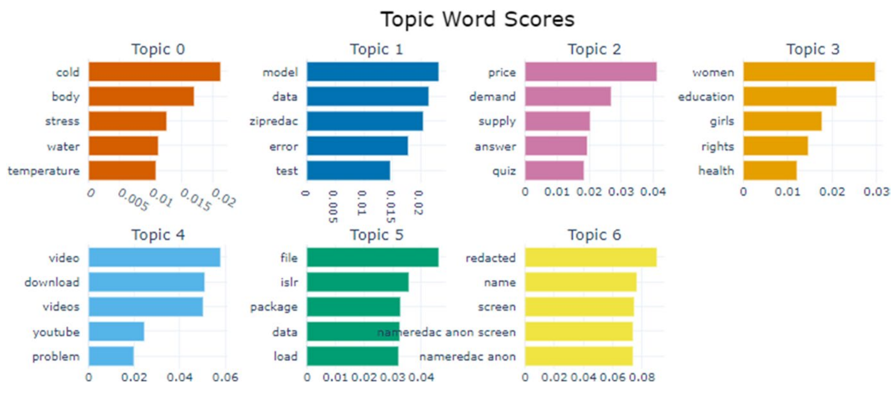

Fig. 4 Topic word scores for Dataset group C

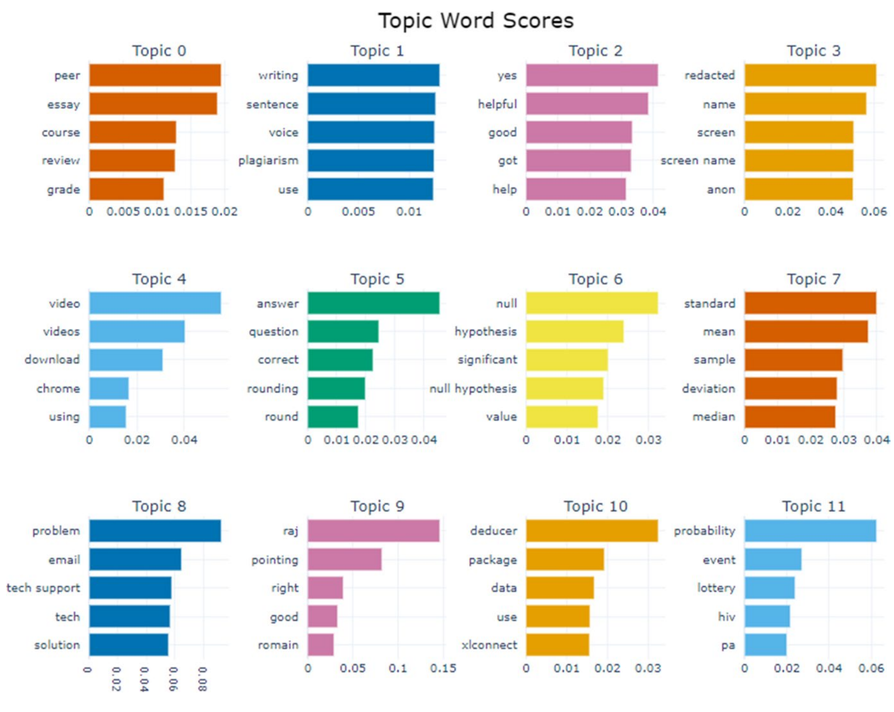

Fig. 5 Topic word scores for Dataset group D

which display the visualizations of the topic word scores for each respective dataset group.

4 BERTopic Comparative analysis with traditional models

We comprehensively evaluated and compared BERTopic and traditional topic models, mainly LDA, LSI, and NMF. The experiments were applied across the four dataset groups: A, B, C, and D. To assess their effectiveness, we utilized two metrics: Coherence metrics score and Inverted Rank-Biased Overlap (RBO) score. Topic Coherence is a metric that measures the quality of an individual topic by quantifying the semantic similarity between highly relevant words within the topic. It helps distinguish between topics that exhibit semantic coherence and those merely statistical artefacts. One commonly used coherence metric is CV, which constructs content vectors based on word co-occurrences and calculates the score using normalized pointwise mutual information (NPMI) and cosine similarity. The coherence score ranges from 0 to 1, with higher values indicating better coherence.

The Inverted Rank-Biased Overlap (IRBO) score evaluates the diversity of topics generated by a particular model. It compares the top N words of two topics and utilizes weighted ranking. A diversity score of 0 suggests redundant topics, while scores closer to 1 indicate greater topic variation. Higher values for both metrics indicate better performance (Murakami, Itsubo, and Kuriyama 2022). By analysing these metrics, we can accurately evaluate the effectiveness of different topic models and identify the models that produce topics with higher levels of coherence and diversity.

The selected approach for determining the optimal number of topics in LDA, LSI, and NMF models involves constructing multiple models with different values for the number of topics (k). The model with the highest coherence score indicates the significance of the generated topics. The process begins with a range of values for the number of topics, typically starting from two and incrementing by two. Only the number of topics is varied as a hyperparameter while keeping other parameters constant until the highest coherence score is achieved. This experiment is conducted across different dataset groups (A, B, C, and D).

Figures 6, 7, 8, and 9 display the coherence scores corresponding to the number of topics for the different dataset groups' LDA, LSI, and NMF models. The optimal number of topics for each model is determined by the highest coherence score, indicating the relevance and significance of the topics. In the case of the BERTopic model, the training process results in 50 topics with a coherence score of 0.616 for dataset group A.

Table 6 presents the optimal number of topics, their corresponding coherence scores for each model for dataset group A, and the IRBO values. For dataset groups B, C, and D, Tables 7, 8, and 9 indicate the optimal numbers of topics determined by the highest coherence scores for the traditional models, and the number of topics automatically obtained using the BERTopic model. These tables also include the IRBO values for evaluation.

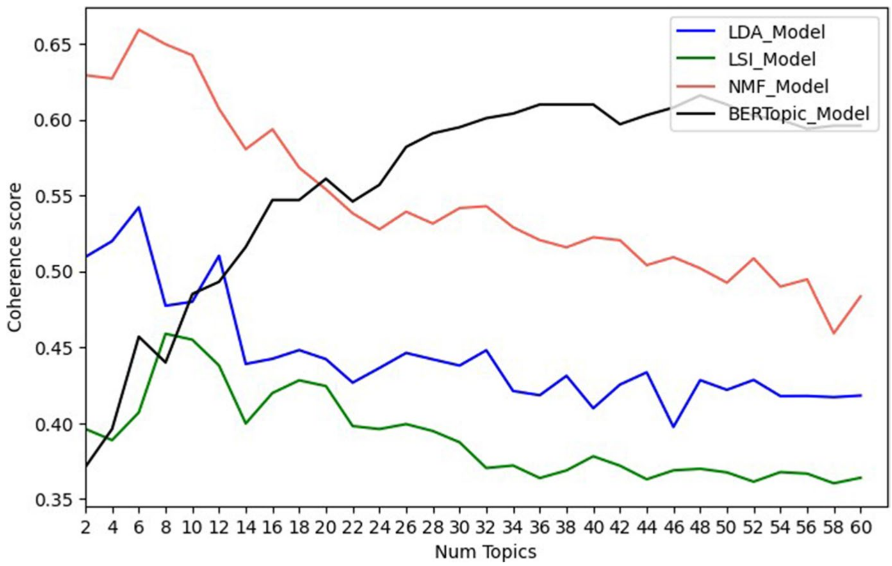

Fig. 6 Optimal number of topics for LDA, LSI, NMF and BERTopic for Group A

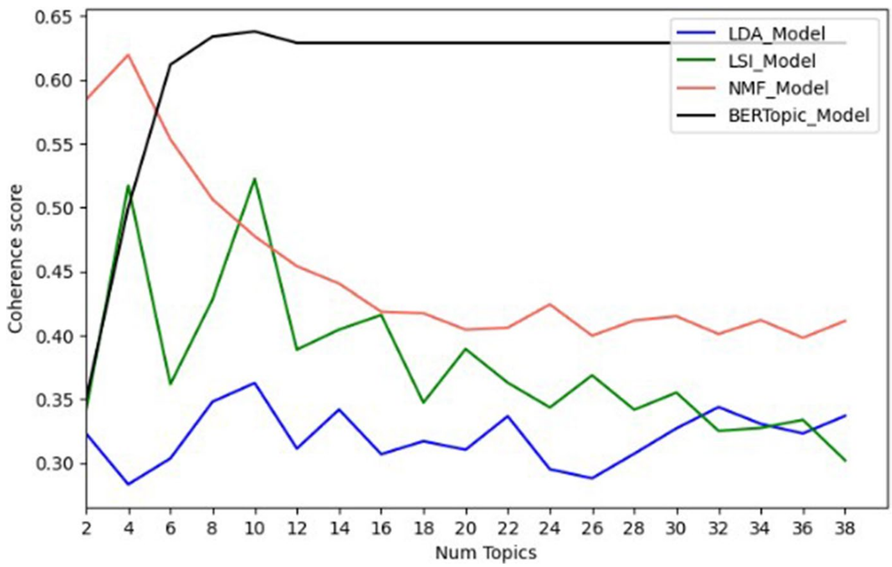

Fig. 7 Optimal number of topics for LDA, LSI, NMF and BERTopic for Group B

The findings in Tables 6, 7, 8 and 9 demonstrate the coherence scores obtained by different topic modeling techniques. For dataset group A, the best coherence score achieved by LDA was 0.542, while the worst was 0.363 for dataset group B. Regarding LSI, the highest coherence score of 0.5117 was observed for

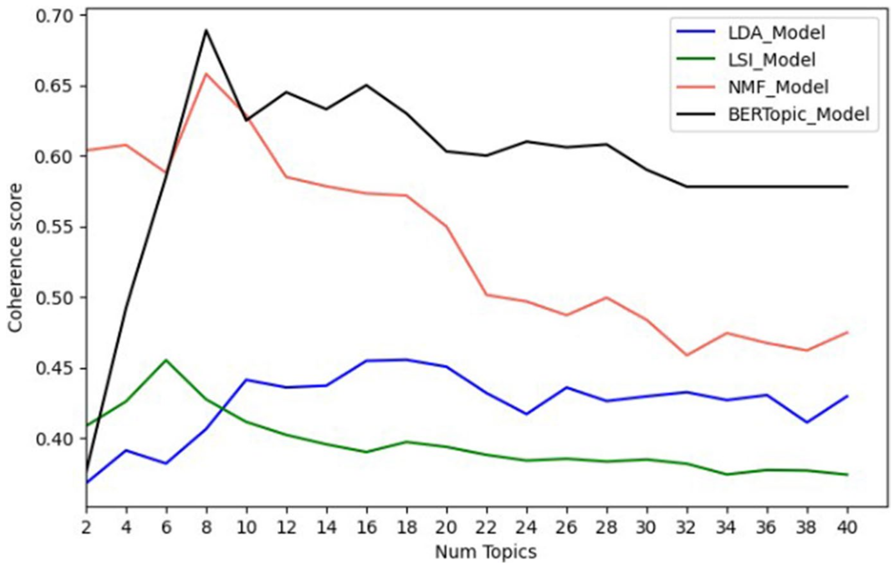

Fig. 8 Optimal number of topics for LDA, LSI, NMF and BERTopic for Group C

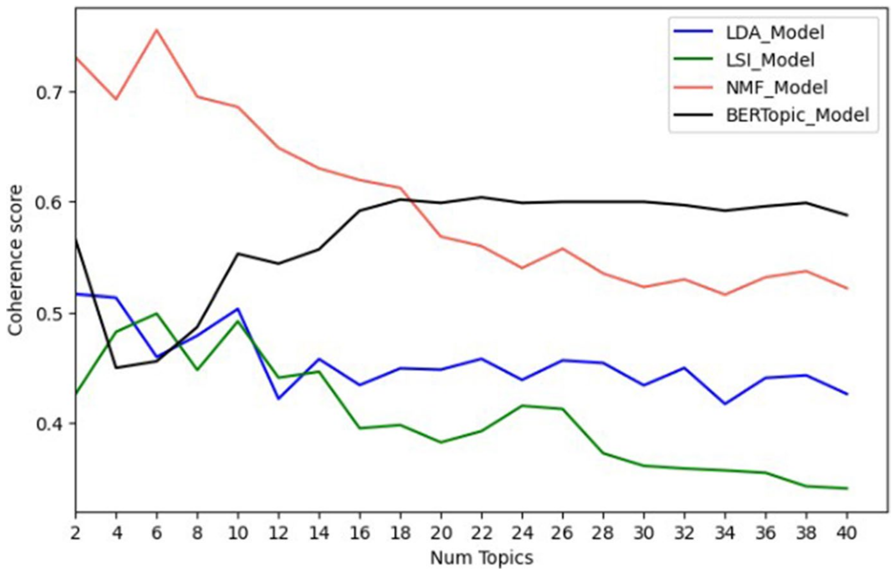

Fig. 9 Optimal number of topics for LDA, LSI, NMF, and BERTopic for Group D

dataset group B, whereas the lowest score of 0.55 was recorded for dataset group C. Regarding NMF, the best coherence score of 0.755 was obtained for dataset group D, while the worst score of 0.620 was observed for dataset group B. Lastly,

Table 6 Optimal number of topics with the corresponding coherence score and IRBO for dataset group A

ALL courses	LDA	LSI	NMF	BERTopic
Number of topics	6	8	6	50
Coherence score	0.542	0.459	0.66	0.616
IRBO	0.974	0.914	1	1

Table 7 Optimal number of topics with the corresponding coherence score and IRBO for dataset group B

Education courses	LDA	LSI	NMF	BERTopic
Number of topics	10	4	4	10
Coherence score	0.363	0.517	0.620	0.638
IRBO	0.724	0.755	0.976	0.98

Table 8 Optimal number of topics with the corresponding coherence score and IRBO for dataset group C

Humanities courses	LDA	LSI	NMF	BERTopic
Number of topics	18	6	8	17
Coherence score	0.455	0.455	0.658	0.689
IRBO	0.972	0.850	0.967	0.99

Table 9 Optimal number of topics with the corresponding coherence score and IRBO for dataset group D

Medicine courses	LDA	LSI	NMF	BERTopic
Number of topics	2	6	6	37
Coherence score	0.517	0.499	0.755	0.604
IRBO	1	0.427	1	1

BERTopic achieved a coherence score of 0.689, the highest for dataset group C, while the lowest score of 0.604 was recorded for dataset group D.

It is evident from these results that NMF and BERTopic models exhibit higher coherence scores compared to LDA and LSI. The BERTopic model achieved coherence scores of 0.616, 0.638, 0.689, 0.611, and 0.604 when utilizing 50, 10, 17, and 37 topics for dataset groups A, B, C, and D, respectively. Similarly, the NMF model obtained coherence scores of 0.66, 0.62, 0.658, and 0.755 when employing 6, 4, 8, and 6 topics for dataset groups A, B, C, and D, respectively. Additionally, both models demonstrated nearly identical high scores close to one for the IRBO metric.

5 Discussion

This study aims to predict the reasons behind learners' urgent posts on MOOC forums by analyzing the textual content they share. To achieve this goal, a comparison was made between BERTopic and various traditional topic models, including LDA, LSI, and NMF.

For training LDA, LSI, and NMF models, a bag-of-words (BoW) corpus was utilized, while BERTopic employed a pre-trained sentence transformer model known as "all-MiniLM-L6-v2". The quality and relevance of the generated course topics across the dataset were evaluated using coherence and RBO scores.

To assess different topic modeling approaches, the reference dataset consisted of student postings labeled as urgent. These urgent postings represented various course types, making the reference dataset representative of various courses. Consequently, experiments were conducted on different subsets of the dataset, with the dataset being divided into four groups: A, B, C, and D, based on the course types they corresponded to.

The LDA, LSI, NMF, and BERTopic models were applied to all dataset groups, yielding notable results. Group A's coherence scores were 0.542, 0.459, 0.66, and 0.616, respectively, while the RBO scores were 0.974, 0.914, 1, and 1. Similarly, the other subset dataset groups exhibited promising coherence and RBO scores, as indicated in Tables 6,7,8, and 9.

Upon examining the graphs presented in Figs. 6, 7, 8, and 9 the results provided in Tables 6,7,8, and 9, it becomes evident that the NMF and BERTopic models exhibited superior performance compared to the other models. The NMF model demonstrated better performance with a small number of topics, whereas the BERTopic model automatically generating the best number of topics with higher levels of coherence.

It is observed from the results that the number of topics generated automatically by the BERTopic model is generally greater than the optimal number of topics produced by other models. The results obtained demonstrated the ability of BERTopic to distinguish and extract diverse topics comprehensively and coherently, facilitating the identification of various reasons behind MOOC forum posts. A larger number of topics indicates a higher level of specificity, as the model considers more detailed information within the data context. Conversely, a smaller number of topics suggests a reduced level of detail.

We employed the trained BERTopic model to identify the primary topics in urgent posts. Table 5 provides illustrative examples of the topics commonly found in learners' urgent inquiries. Leveraging the trained BERTopic model for topic prediction can substantially improve the responsiveness and relevance of a chatbot specifically designed to address urgent posts.

While BERTopic models demonstrate superior performance, there are potential limitations to consider. Firstly, the computational demands of BERT models can pose challenges for users with limited computational resources or when handling large datasets. Secondly, biases in language understanding may arise, as BERT models can inherit biases from their training data, potentially skewing topic representations. Additionally, the scalability of BERTopic in environments with limited computational resources, such as MOOC platforms with thousands of concurrent users, needs to be addressed. A statistical significance analysis between model performances could further enhance the study.

6 Conclusion

MOOCs face challenges such as low completion rates and high dropout rates. The limited number of instructors compared to many learners contributes to a lack of interaction and timely response to urgent posts from learners. Previous research on urgent posts has focused on identifying them without delving into their underlying causes. This study, however, breaks new ground by examining the causes of urgent posts using topic modeling techniques to predict their root causes automatically.

To accomplish this, the researchers employed four well-established models: LDA, LSI, NMF with BOW corpora, and BERTopic with sentence embeddings. Before training the models, a natural language processing (NLP) pipeline is applied for data cleaning and preprocessing. The models were evaluated using coherence and RBO scores, assessing the generated topics' quality and diversity.

Based on the experimental results, it was observed that the NMF and BERTopic models achieved better performance compared to the other models. The NMF model exhibited superior results when a smaller number of topics was utilized, whereas the BERTopic model excelled in generating topics with higher levels of coherence when a larger number of topics was required.

Automatic classification of urgent posts and automatic prediction of their causes help to deal with those causes more professionally and specialized. This facilitates the instructors of a recommendation system that considers learners' problems and then directs course administrators to address them. Or build a chatbot to help students solve their urgent problems personally. In addition, we aspire to evaluate our approach to MOOCs to highlight the multiple problems and their causes and deal with them automatically in MOOCs.

The automated classification of urgent posts and the subsequent prediction of their causes contribute to a more professional and specialized approach to addressing these underlying issues. This, in turn, facilitates the implementation of various solutions in the context of MOOCs. For instance, it enables the development of a recommendation system that considers learners' problems and directs course administrators to address them effectively. Alternatively, it allows for the creation of a chatbot that can personally assist students in resolving their urgent problems.

Furthermore, our approach to evaluating MOOCs aims to shed light on the multiple problems and their respective causes and to automate the process of addressing them within the MOOC environment. By employing automated techniques, we aspire to enhance the overall experience of learners and instructors by efficiently identifying and resolving these issues promptly.

7 Future work

While the study has provided insights into topic modeling in the context of MOOC platforms, with the aim of investigating the reasons behind urgent posts in MOOC forums, there are several potential directions for future research.

These directions aim to expand and improve the current understanding of topic modeling techniques Large Language Models (LLMs), as well as validate the generalizability of the results across different platforms and developing new evaluation metrics that can assess the generated topics.

Fine-tuning LLMs for MOOC-specific tasks holds significant promise. Recent research trends emphasize the need to adapt LLMs effectively to enhance learning outcomes. For instance, (Wei et al. 2023) explored the use of deep-learning-based natural language generation (NLG) models to provide personalized support in MOOC discussion forums, bridging the gap between human-generated and automated responses.

To verify the generality of the results obtained from the current study, it is essential to extend the research to other MOOC platforms. While the findings from the current study provide insights into topic modeling within a specific platform, the landscape of MOOC platforms is diverse, with variations in course topics, learner demographics, and platform features. Conducting similar studies on different MOOC platforms can help evaluate the generalizability of the identified topics and uncover platform-specific variations. Moreover, comparing the results across multiple platforms will contribute to a more comprehensive understanding of the topics prevalent in the MOOC domain. In addition, conducting comparative studies with other educational platforms, such as Learning Management Systems (LMS) or online discussion boards in traditional classrooms, can be beneficial.

Another aspect that warrants attention in future research is the evaluation of topic modeling performance. While the current study focused on topic extraction and interpretation, it is important to develop robust evaluation metrics that can quantitatively assess the quality and coherence of the generated topics. Exploring new evaluation techniques and metrics specific to MOOC platforms can contribute to a more rigorous assessment of topic modeling algorithms.

By exploring these future research directions, the field can advance our understanding of the underlying reasons behind urgent posts in MOOC forums, leading to more effective interventions, enhanced learner support, and improved course completion rates.

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Data availability We conducted our experiments using the Stanford MOOC Posts dataset (A. Agrawal and Paepcke 2014). It is available on demand at the link https://datastage.stanford.edu/StanfordMoocPosts/.

Declarations

Conflict of interests None.

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Authors and Affiliations

Nabila Khodeir[1] · **Fatma Elghannam**[1]

Nabila Khodeir
nkhodeir@eri.sci.eg

Fatma Elghannam
elghannam@eri.sci.eg

[1] Informatics Research Department, Electronics Research Institute, Elbahth Elemy, Cairo 11843, Egypt